

Ramiro M. Irastorza | Dr. Eng. Independent Researcher (CONICET)

IFLySiB (CONICET-UNLP) – La Plata – Argentina

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Education

National University of La Plata (UNLP)

Argentina

Ph.D. in Electronic Engineering

2010

Dissertation: *System Identification: Application to the In Vitro Evaluation of Bone Quality in Human Trabecular Tissue.*

National University of La Plata (UNLP)

Argentina

Electronic Engineer

2005

Current Positions

CONICET

Independent Researcher

2026–Present

Automation Laboratory, UTN-FRLP

Mechanical Engineering Dept.

Full Professor

2025–Present

Management: Vice Director, Center for Experimental and Computational Mechanics (CIMEC), UTN-FRLP.

Research Interests

Computational bioengineering, Finite element modeling, Microwave tomography, Radiofrequency medical therapies, Scientific computing and machine learning.

Research Funding and Leadership

UTN Project

Principal Investigator

2023–2026

Biomedical and Biomimetic Problems Solved Using Finite Elements.

National Research Project (PICT - Agency)

Principal Investigator

2022–2024

Microwave Tomography: Reconstruction Algorithms, Experimental Validation and Applications.

CONICET Project

Co-Principal Investigator

2023–2024

Inverse Problems in Biomedical Engineering Using Sparse Representations and AI.

Selected International Projects

2023–2025: Advanced Ablative Therapies (Spain): Funded by Ministry of Economy and Competitiveness. Collaboration between UPV (Spain), IFLySiB, LEICI, and ENyS.

2015–2017: Minimally Invasive Surgery (Spain): Funded by Ministry of Economy and Competitiveness. Collaboration between UPV and IFLySiB.

2010–2012: AUF International Lab: Collaboration between Université Laval (Canada), INTI, and UBA (Argentina).

Graduate Teaching

UTN-FRLP: Finite Element Method with Open-Source Software

UNLP / UTN: Computational Tools for Scientists

UNAJ: Statistical Methods / Applied Mathematics Complements

UPV (Spain): Radiofrequency and Biological Tissues

Human Resources Training

- Supervision of Ph.D. students in biomedical engineering and microwave imaging.
- Advisor of undergraduate and graduate theses at UTN, UNAJ, and UNLP.
- Supervisor of CONICET and national research fellowship recipients.

Selected Publications

2025: Berjano E., **Irastorza R.M.**, *Positioning of the dispersive electrode and its effect on the safety and efficacy of radiofrequency ablation*, EP Europace.

2024: Collavini S., et al., **Irastorza R.M.**, *Impact of surrounding tissue-type and peri-electrode gap in SEEG guided radiofrequency thermocoagulation*, Int. J. Hyperthermia.

2023: **Irastorza R.M.**, et al., *Anterior vs. posterior position of dispersive patch during RF catheter ablation: in silico modelling*, EP Europace.

2022: González-Suárez A., et al., **Irastorza R.M.**, *Computer modeling of radiofrequency cardiac ablation: 30 years of bioengineering research*, CMPB.

2021: Madrid M.A., **Irastorza R.M.**, Meyra A.G., Carlevaro C.M., *Self-assembly of self-propelled magnetic grains*, EPJ Web of Conferences.

2020: **Irastorza R.M.**, Gonzalez-Suarez A., Pérez J.J., Berjano E., *Differences in applied electrical power between full thorax models and limited-domain models for RF cardiac ablation*, IJH.

2019: Fajardo J.E., et al., **Irastorza R.M.**, *Microwave Tomography with Phaseless Data on the Calcaneus by Means of ANN*, Med & Biol Eng & Comp.

2019: Sánchez H. R., **Irastorza R.M.**, Carlevaro C.M., *Uncertainties and temperature correction in molecular dynamic simulations of dielectric properties of condensed polar systems*, Journal of Molecular Liquids.

Professional Activities

Reviewer: Int. J. Hyperthermia, Med. Eng. & Physics, PIER, Int. J. Num. Meth. Biomed. Eng.

Guest Editor: Journal of Cardiovascular Development and Disease (Special Issue on Cardiac Ablation).

Coordination: Co-Coordinator of IT, Communications and Electronics Area, FONCyT (2022–2025).